ELEC70004
EE9-AC6

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE PART IV: MEng and ACGI

ANALOGUE SIGNAL PROCESSING

Friday, 16 May 2025, 10:00 AM

Time allowed: 3:00 hours

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : E. Rodriguez-Villegas

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Special instructions for students

Unless otherwise stated the following parameters have the following definitions:

V_{DS} *: Drain Source Voltage.*

V_{GS} *: Gate Source Voltage.*

V_{TH} *: Threshold Voltage.*

U_t *: Thermal Voltage.*

n *: Weak Inversion Slope factor.*

g_m*: Transconductance.*

1.

a. Compare Voltage and Current mode domains as regards to dynamic range, signal processing and speed.

[4]

b. The circuit shown in Figure 1.1 has MOS transistors which are operating in the weak-inversion region of operation with a slope factor n = 1.23.

i) Show that the output current $I_{out} = I_1 - I_2$ of the circuit may be given by equation 1. Show all your steps in derivation including the derivation of currents I_1 and I_2.

$$I_{out} = I_{DC} \sinh\left(\frac{V_1 - V_2}{nU_t}\right) \quad (1)$$

[4]

ii) Derive the small signal transconductance and thus derive the transconductance efficiency of the circuit in Figure 1.1 stating all your assumptions. You may use the Taylor series expansion $coshx = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots$ in your analysis.

[4]

c. For the cascade of amplifiers shown in Figure 1.2, derive the output referred noise and explain what factors can be optimized to improve the signal to noise ratio.

[8]

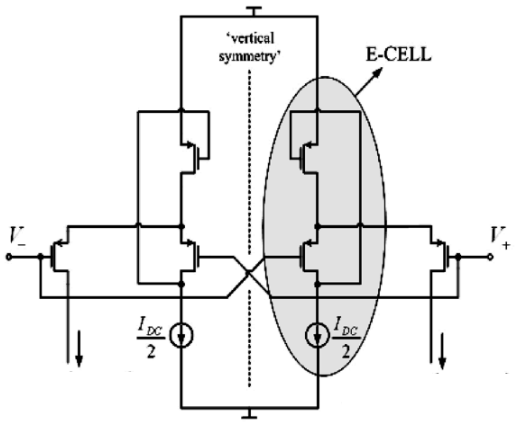


Figure 1.1

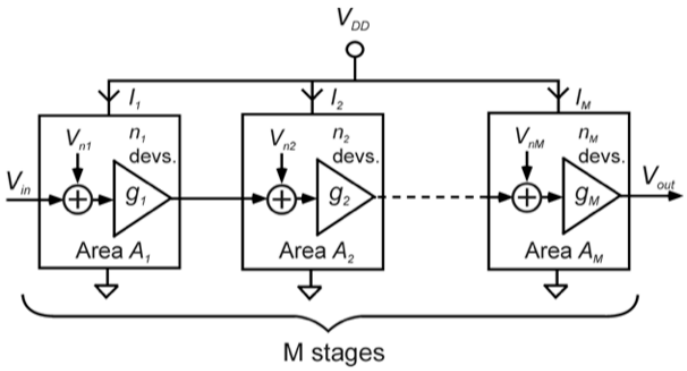


Figure 1.2

2.

a. State and derive the translinear principle (TLP) for a loop of MOS transistors working in weak inversion. Give all assumptions you make.

[5]

b. Show how short channel effects in MOS transistors operating in weak inversion lead to multiplicative errors in output currents of translinear circuits.

[4]

c. Figure 2.1 shows a translinear circuit.

i. Derive the transfer function I_{out} of this circuit using the translinear principle. You may assume that currents I_1 and I_3 are input currents and I_4 and I_2 are a static bias'.

[5]

ii. Write down an expression for the output current I_{out} when I= Asin(ωt) and the input currents I_1 and I_3 are given as:

$$I_3 = \left|\frac{dI}{dt}\right|, \qquad I_1 = \left|\int I \quad dt\right|$$

[4]

iii. What is the function of this circuit when the inputs are applied as in 2.c.ii ?

[2]

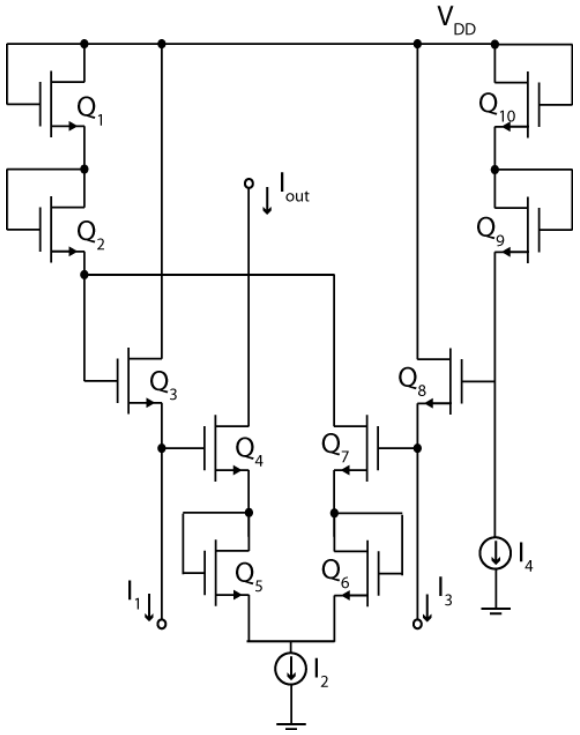


Figure 2.1

3.

a) Figure 3.1 shows a block diagram of a current-mode companding integrator, with an input X and an output Y. f() and g() represent non-linear functions which compress and expand the signal such that the input-output relationship of the integrator is linear and given by:

$$Y = \tau \int X \cdot dt$$

Given that the expansive function is defined as Iout = g(Vc) = $I_0\sinh(V_c/nU_t)$, derive the function f(x) such that linear integration is achieved. All constants should be grouped to represent a current I_1, with the function f(x) composed of just I_{in}, I_1 and I_{cosh}, whereby $I_{cosh} = I_0\cosh(V_c/nU_t)$.

[6]

b) The transfer function of a second order filter topology has been decomposed into the following state-space equations:

$$\dot{X_1} = -\left(\frac{\omega_0}{2Q}\right)X_1 - \omega_0\left(1 - \frac{1}{4Q^2}\right)X_2 + \omega_0 U$$
$$\dot{X_2} = -\left(\frac{\omega_0}{2Q}\right)X_2 + \omega_0 X_1$$
$$Y_1 = X_1$$
$$Y_2 = X_2$$

whereby Y_1 and Y_2 are the outputs, X_1 and X_2 are the state-variables and U is the input.

i) Show that the output Y_1 can be used to implement a second order lowpass transfer function and the output Y_2 can be used to implement a "two-pole one-zero" second order transfer function.

[4]

ii) By using the mappings below, show how these state space equations can be mapped to non-linear log-domain design equations. State any assumptions you make.

$$X_1 = I_1 \exp\left(\frac{V_1}{nU_t}\right) \qquad X_2 = I_2 \exp\left(\frac{V_2}{nU_t}\right) \qquad U = I_U \exp\left(\frac{V_U}{nU_t}\right)$$

[5]

iii) With these log-domain design equations, sketch a schematic of the final log domain topology which realises the two outputs Y_1 and Y_2 using weak inversion MOS transistors.

[5]

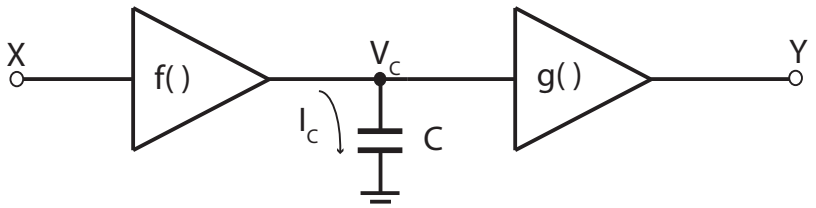


Figure 3.1

4.

a) Figure 4.1 shows a conventional second-generation positive current conveyor, CCII+.

i) Explain its operation principle and describe how impedances at each of its three ports, X, Y, Z, differ from a standard operational amplifier.

[4]

ii) Draw the circuit of a transistor implementation of a bi-directional CCII+ which does not use any opamps and explain its operation.

[4]

iii) Using current conveyors design circuits which implement:

- A grounded inductance.
- A full wave voltage mode rectifier.

[4]

b) Derive the generalized equation for the output currents, I_{outi} of the circuit in Figure 4.2 and explain the function of this circuit.

[4]

c) Derive the output current, I_{out} for the switch current algorithmic cell given in Figure 4.3 showing how this eliminates any error caused due to the switches.

[4]

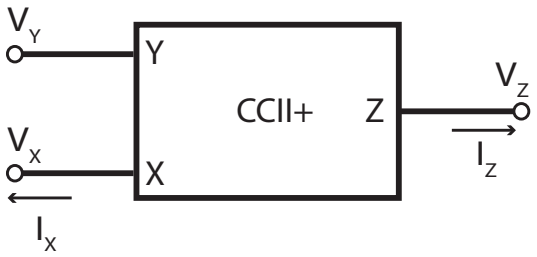


Figure 4.1

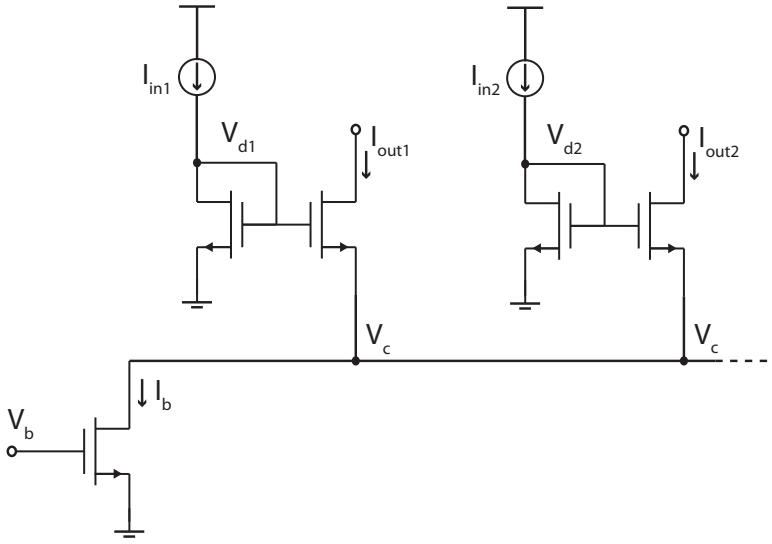


Figure 4.2

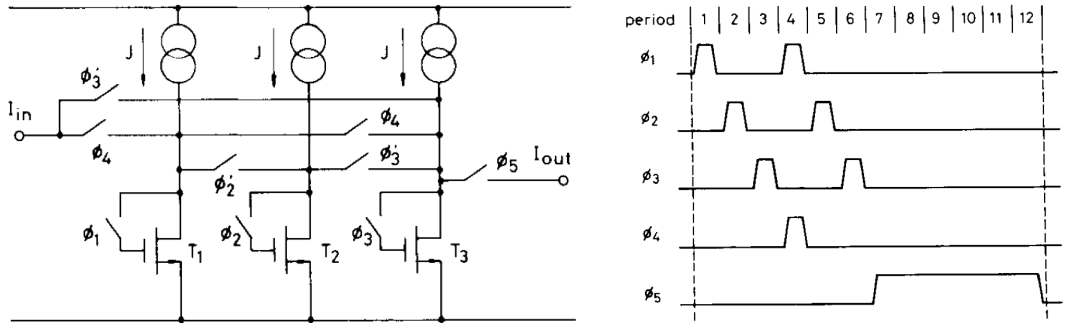

J
J
J
ϕ_3
I_{in}
ϕ_4
ϕ_4
ϕ_2
ϕ_3
ϕ_5
I_{out}
ϕ_1
T_1
ϕ_2
T_2
ϕ_3
T_3
period
1
2
3
4
5
6
7
8
9
10
11
12
ϕ_1
ϕ_2
ϕ_3
ϕ_4
ϕ_5

Figure 4.3

5. This question is related to switched capacitor circuits.

a) State four non-idealities of using MOSFETs as switches in switched capacitor circuits and write the equation for the change in voltage, ΔV_{total}, considering all four factors.

[4]

b) For the most dominant non-ideal factor from Q5(a), explain using equations and diagrams how the output deviates from the ideal input, considering the body effect.

[4]

c) Define a Figure of Merit (FoM) that considers the trade-off in speed and precision, with only first order effects. Based on the result, explain how you would design the size of the switch transistor using an n-MOSFET in a 180nm CMOS process to achieve a good FoM.

[3]

d) State the trade-offs with respect to gain, precision and speed between Figure 5.1 (a) and Figure 5.1 (b)

[3]

e) Based on the conclusion of Q5(d), design a precision multiply-by-3 circuit. Explain its operating principle and describe how it achieves better precision compared to the equivalent of the circuit in Figure 5.1 (b) when C1 = 3 x C2.

[4]

f) Redraw the circuit in Figure 5.1(b) with polarised capacitors used in CMOS processes, as illustrated in Figure 5.2. Explain your choice of polarisation.

[2]

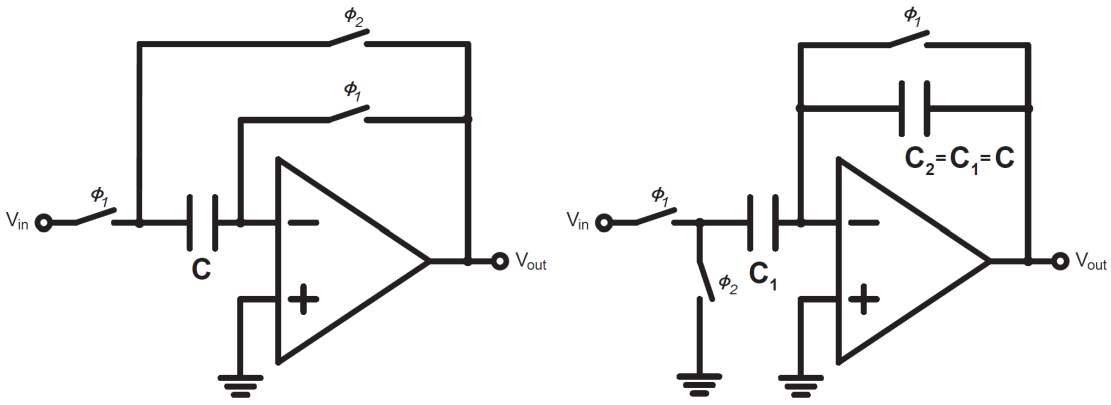


Figure 5.1

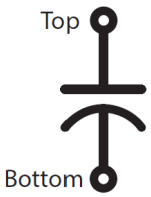


Figure 5.2

6. This question is related to a sensor instrumentation system using switch capacitor techniques.

a) Design a closed-loop auto-zeroing circuit using auxiliary compensation for a fully differential (differential input and differential output) 2-stage amplifier, and state the residual offset seen at the input of the main amplifier.

[4]

b) State the function of the two circuits shown in Figure 6.1 and state the main benefit of the scheme shown in Figure 6.1(b).

[2]

c) Analyse the circuit in shown Figure 6.2 deriving its output voltage at each phase of operation and state its functionality.

[4]

d) (i) Figure 6.3 shows a typical analogue signal processing chain that continuously amplifies an output signal from a differentially configured sensor pair to a differential input ADC. You are responsible for designing a signal processing pipeline using the following criteria:

- The sensor's output has a frequency range of 1–10 Hz, and a signal magnitude of 10 mV. Each op-amp has a 1/f noise amplitude of 10 mV at 10 Hz, and a corner frequency of 1 kHz (you don't need to show the circuit inside the opamp).
- You should design the circuit so that the sensor's output is not buried by the input-referred noise of your circuit.
- You must adapt the circuit in Figure 6.2 as one of your differential blocks in the pipeline.

[4]

(ii) Explain the method used to achieve a good noise performance in your sensing system. You may use frequency spectrum plots to aid your description.

[4]

(iii) State the effect at the input of the ADC due to a finite bandwidth of the main amplifier.

[2]

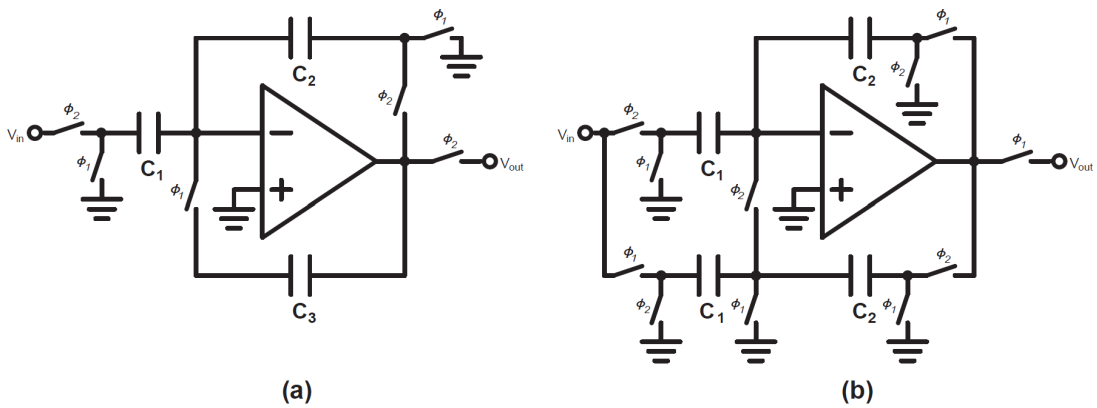


Figure 6.1

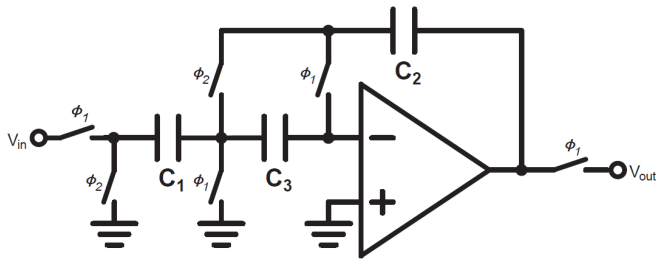


Figure 6.2

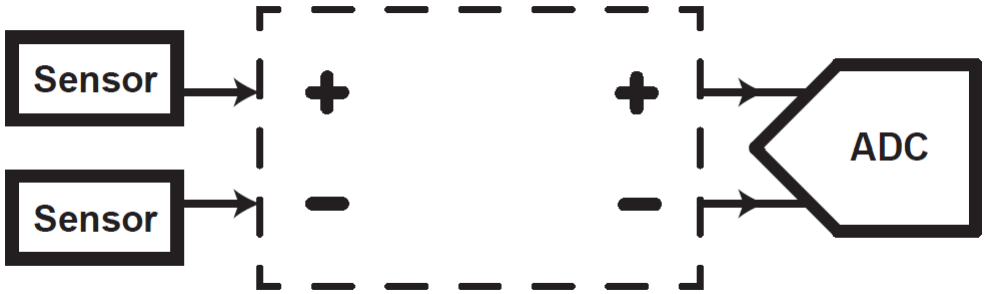


Figure 6.3